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SUBJECT: Lab 9, Measurement Statistics

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1.0 Statement of Purpose

The purpose of this experiment was to measure the distance of 100 nerf darts that were thrown, where we only considered horizontal displacement. From this we measured variability through a data analysis, by finding the standard deviation, mean, and relative frequency distribution. These were used to find error distribution throughout the trials to see if it follows the behavior of a typical error distribution function.

2.1 Background

For this experiment, multiple recorded measurements, it is essential to analyze the measurement error as the data set will include variability within the set. Conducting a statistical analysis allows us to have a framework to base our conclusions on any claims that are made.

The standard deviation measures the spread of measurements based on the mean of the data set. If a small standard deviation is calculated, the data set is precise with small uncertainty values. If a large standard deviation is calculated, then we can see that the measurements have large ranges of uncertainty, which means the data is not reliable nor should it follow the normal error curve. The equation for Standard deviation is shown below:

$$\sigma = \sqrt{\frac{1}{n} \sum (x_i - x_{mean})^2}$$

For the relative frequency distribution, it is obtained through dividing the number of occurrences from the data set for each bin by the n amount of measurements(100 nerf shots). To obtain the normal error distribution function for this data set, the bin width is needed to produce the relative frequency distribution function.

The equation to plot the normal error distribution function is shown below:

$$P(x) = \left[\frac{1}{\sigma\sqrt{2\pi}} \right] \exp \left[-\frac{(x_i - x_{mean})^2}{2\sigma^2} \right]$$

3.0 Discussion

3.1 Equipment

- Nerf Gun
- Nerf Darts
- Whiteboard with markings
- Laptop

3.2 Procedure

For this lab, the experiment involved shooting 100 Nerf darts with a Nerf gun to a centerline and measuring the variation (horizontal distance) from the centerline. We fired 100 Nerf darts with the same 8 Nerf Darts and obtained a mean of 2.495cm. The measurements were analyzed and computed on Excel, and from this we conducted data analysis to obtain a normal error curve and relative frequency distribution function. With our data, it would be possible to see if our measurements follow the behavior of a normal error curve.

3.3 Experimental Data

The following Table shows the data set from all 100 Nerf dart shots, with the upper bin limit, lower bin limit, number of occurrences, and the frequency distance. The data set had a mean of 2.495 cm and a standard deviation of 5.32 cm. The following table below has listed all the data that was used in Excel, which was used to create the later graphs.

Table 1. Raw data showing all data analysis and 100 measurements

Locations, cm	x - xm	(x - xm)^2		Lower bin limit, cm	Upper Bin Limit, cm	Bin Center, cm	Number of Occurrences	fraction of total number	Normal error curve
-14.0	-16.495	272.0850		-18.5	-17.5	-18	0	0.00	0.00004
-10.0	-12.495	156.1250		-17.5	-16.5	-17	0	0.00	0.00009
-9.0	-11.495	132.1350		-16.5	-15.5	-16	0	0.00	0.00017
-8.0	-10.495	110.1450		-15.5	-14.5	-15	0	0.00	0.00032
-7.0	-9.495	90.1550		-14.5	-13.5	-14	1	0.01	0.00059
-6.0	-8.495	72.1650		-13.5	-12.5	-13	0	0.00	0.00104
-6.0	-8.495	72.1650		-12.5	-11.5	-12	0	0.00	0.00177
-5.0	-7.495	56.1750		-11.5	-10.5	-11	0	0.00	0.00292
-4.0	-6.495	42.1850		-10.5	-9.5	-10	1	0.01	0.00465
-4.0	-6.495	42.1850		-9.5	-8.5	-9	1	0.01	0.00713
-4.0	-6.495	42.1850		-8.5	-7.5	-8	1	0.01	0.01056
-4.0	-6.495	42.1850		-7.5	-6.5	-7	1	0.01	0.01508
-4.0	-6.495	42.1850		-6.5	-5.5	-6	2	0.02	0.02079
-3.0	-5.495	30.1950		-5.5	-4.5	-5	1	0.01	0.02766
-3.0	-5.495	30.1950		-4.5	-3.5	-4	5	0.05	0.03550
-3.0	-5.495	30.1950		-3.5	-2.5	-3	4	0.04	0.04397

-3.0	-5.495	30.1950		-2.5	-1.5	-2	4	0.04	0.05255
-2.0	-4.495	20.2050		-1.5	-0.5	-1	5	0.05	0.06060
-2.0	-4.495	20.2050		-0.5	0.5	0	9	0.09	0.06744
-2.0	-4.495	20.2050		0.5	1.5	1	6	0.06	0.07242
-2.0	-4.495	20.2050		1.5	2.5	2	2	0.02	0.07503
-1.5	-3.995	15.9600		2.5	3.5	3	11	0.11	0.07502
-1.0	-3.495	12.2150		3.5	4.5	4	10	0.10	0.07238
-1.0	-3.495	12.2150		4.5	5.5	5	7	0.07	0.06738
-1.0	-3.495	12.2150		5.5	6.5	6	8	0.08	0.06053
-1.0	-3.495	12.2150		6.5	7.5	7	6	0.06	0.05247
-0.5	-2.995	8.9700		7.5	8.5	8	3	0.03	0.04389
0.0	-2.495	6.2250		8.5	9.5	9	3	0.03	0.03542
0.0	-2.495	6.2250		9.5	10.5	10	4	0.04	0.02759
0.0	-2.495	6.2250		10.5	11.5	11	1	0.01	0.02073
0.0	-2.495	6.2250		11.5	12.5	12	0	0.00	0.01503
0.0	-2.495	6.2250		12.5	13.5	13	1	0.01	0.01052
0.0	-2.495	6.2250		13.5	14.5	14	3	0.03	0.00710
0.0	-2.495	6.2250		14.5	15.5	15	0	0.00	0.00463
0.0	-2.495	6.2250		15.5	16.5	16	0	0.00	0.00291
0.5	-1.995	3.9800		16.5	17.5	17	0	0.00	0.00177
1.0	-1.495	2.2350		17.5	18.5	18	0	0.00	0.00103
1.0	-1.495	2.2350				Totals	100	1.00	
1.0	-1.495	2.2350							
1.0	-1.495	2.2350							
1.0	-1.495	2.2350							
2.0	-0.495	0.2450							
2.0	-0.495	0.2450							
2.5	0.005	0.0000							
3.0	0.505	0.2550							
3.0	0.505	0.2550							
3.0	0.505	0.2550							
3.0	0.505	0.2550							
3.0	0.505	0.2550							
3.0	0.505	0.2550							
3.0	0.505	0.2550							
3.0	0.505	0.2550							
3.0	0.505	0.2550							
3.5	1.005	1.0100							
3.5	1.005	1.0100							
4.0	1.505	2.2650							
4.0	1.505	2.2650							
4.0	1.505	2.2650							

4.0	1.505	2.2650							
4.0	1.505	2.2650							
4.0	1.505	2.2650							
4.0	1.505	2.2650							
4.0	1.505	2.2650							
4.5	2.005	4.0200							
4.5	2.005	4.0200							
5.0	2.505	6.2750							
5.0	2.505	6.2750							
5.0	2.505	6.2750							
5.0	2.505	6.2750							
5.0	2.505	6.2750							
5.5	3.005	9.0300							
6.0	3.505	12.2850							
6.0	3.505	12.2850							
6.0	3.505	12.2850							
6.0	3.505	12.2850							
6.0	3.505	12.2850							
6.0	3.505	12.2850							
6.0	3.505	12.2850							
6.0	3.505	12.2850							
7.0	4.505	20.2950							
7.0	4.505	20.2950							
7.0	4.505	20.2950							
7.0	4.505	20.2950							
7.0	4.505	20.2950							
7.0	4.505	20.2950							
7.5	5.005	25.0500							
8.0	5.505	30.3050							
8.0	5.505	30.3050							
9.0	6.505	42.3150							
9.0	6.505	42.3150							
9.0	6.505	42.3150							
9.5	7.005	49.0700							
10.0	7.505	56.3250							
10.0	7.505	56.3250							
10.0	7.505	56.3250							
11.0	8.505	72.3350							
13.0	10.505	110.3550							
14.0	11.505	132.3650							
14.0	11.505	132.3650							
14.0	11.505	132.3650							
Mean	STDEV	(x-x)^2							
2.495000	5.320294	2802.247500							

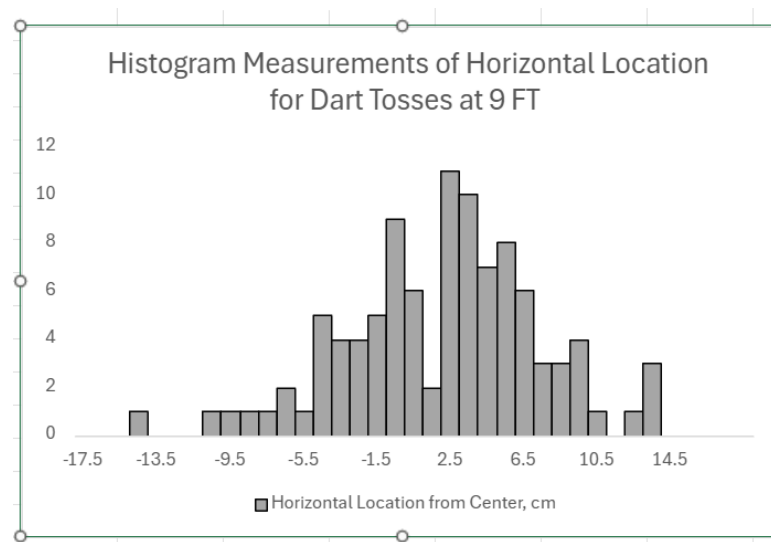


Figure 1. Histogram of 100 horizontal darts. bin width was set at 1cm

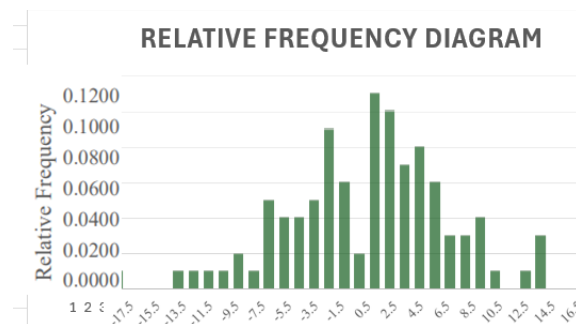


Figure 2. Relative frequency diagram of 100 horizontal darts, with 1cm bin width, showing the number of occurrences

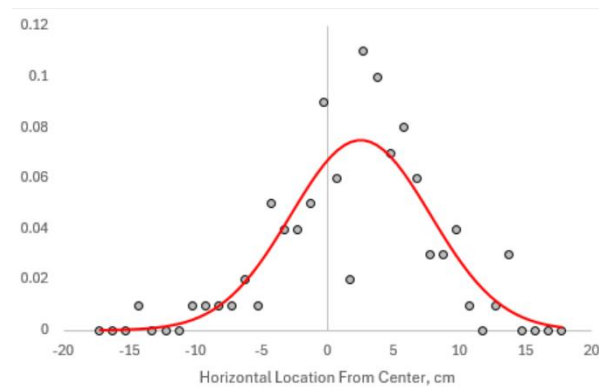


Figure 3. Relative frequency distribution function next to the normal error curve. Behavior shows similar to curve and function.

3.4 Interpretation

The normalized variable that we used in this lab is listed below. This equation is what was used to see the fraction of value from -9.5 to 9.5cm. The fraction of values is 89%

$$\eta = \frac{x - x_m}{\sigma}$$

$$\frac{-9.5 - 2.495}{5.32} = 2.26$$

$$\frac{9.5 - 2.495}{5.32} = 1.32$$

$$P(x) = \left[\frac{1}{\sigma\sqrt{2\pi}} \right] \exp \left[-\frac{(x_i - x_{mean})^2}{2\sigma^2} \right]$$

$$P(1.32) = .4066$$

$$P(2.26) = .4881$$

$$0.4066 + .4881 = 0.894 * 100\% = 89\%$$

From the fraction of value from -9.5 to 9.5, we can understand that 89% of the 100 darts were thrown within the range of -9.5 to 9.5 cm from the centerline(normal error curve). Since 13 of our points from figure 3. Are outside of the curve, we can get the actual percentage by simply doing 87/100 = 87%, which is pretty close to the predicted 89%.

4.0 Conclusions

The measurements after our data analysis produced a mean of 2.495 cm. Our standard deviation value was 5.32cm ($\pm 2.66cm$). These values let us know that the nerf dart data points were shot with a moderate spread. In regard to the normal distribution, it had a predicted fraction of values within $\pm 9.5cm$ to be about 89% of the data set, and our actual was 87%. This makes sense when looking at Figure 3. and seeing the normal error curve next to our data set and seeing most data fall underneath the curve.

The normal error distribution function shows similar behavior that we would have expected to see, which means that the uncertainty and variability from our data points are indeed random. If they were showing deviations or a pattern, that would indicate that our model was not correct and that there is a systemic error causing the pattern, but since this is not the case we know we produced a normal error distribution with our measurements.